Exercise Set 7: Tangent Spaces and Vector Fields

In Set 6 we learned that smooth manifolds allow differentiation. In this set we make that precise: the ***tangent space*** T_pM *at a point* p *is the space of all possible velocities of smooth curves through* p*. It is the linearisation of the manifold at that point. Assembling all tangent spaces gives the* ***tangent bundle*** TM*, and a smooth choice of tangent vector at every point is a* ***vector field****. These are the fundamental objects for writing differential equations on manifolds.*

Recall of Definitions and Examples

Tangent vectors

R1. Write the definition of a **tangent vector at** $p \in M$ using smooth curves. What does it mean for two curves γ_1 and γ_2 to define the same tangent vector?

R2. Write the definition of the **directional derivative**: given $v \in T_pM$ represented by a curve γ with $\gamma(0) = p$, how does v act on a smooth function $f \in C^\infty(M)$?

R3. Write the definition of a **derivation at** p. State (without proof) the theorem that every derivation at p is a tangent vector and vice versa.

R4. In a local chart (U, ϕ) with coordinates $(x^1, \dots, x^d)$, write the **coordinate basis vectors** $\frac{\partial}{\partial x^i}\Big|_p$ and their action on a smooth function f. How many basis vectors are there? What is $\dim T_pM$?

R5. Write the definition of the **pushforward** $df_p : T_pM \to T_{f(p)}N$ of a smooth map $f : M \to N$. How does it act on a tangent vector v represented by a curve γ? Write the formula.

Vector fields

R6. Write the definition of a **vector field** $X \in \mathfrak{X}(M)$ as a section of the tangent bundle. What condition does $\pi \circ X = \mathrm{id}_M$ enforce? Why is this condition necessary?

R7. In local coordinates $(x^1, \dots, x^d)$, every vector field writes as

$$X\Big|_U = \sum_{i=1}^{d} X^i(x)\, \frac{\partial}{\partial x^i}.$$

What are the $X^i(x)$? What does smoothness of X mean in terms of the X^i?

R8. Write the definition of the **Lie bracket** $[X, Y]$ of two vector fields. Give the local coordinate formula. What does $[X, Y] = 0$ mean geometrically in terms of flows?

R9. For each of the following, state whether it is a vector field on the given manifold. If not, explain why.

(a) $X(x, y) = (x, y)$ on $\mathbb{R}^2$

(b) The outward unit normal on $S^2 \subset \mathbb{R}^3$

(c) $X(\cos\theta, \sin\theta) = (-\sin\theta, \cos\theta)$ on S^1

(d) A nowhere-vanishing vector field on S^2

Computation

Tangent vectors at a point

C1. (Tangent vectors on S^1.)

Let $p = (1, 0) \in S^1$ and consider the following curves:

$$\gamma_1(t) = (\cos t, \sin t), \qquad \gamma_2(t) = (\cos 2t, \sin 2t), \qquad \gamma_3(t) = (\cos t^3, \sin t^3).$$

All satisfy $\gamma_i(0) = p$.

(a) Compute the tangent vector $v_i = \gamma_i'(0)$ for each curve as an element of $T_p\mathbb{R}^2$.

(b) Which of v_1, v_2, v_3 lie in T_pS^1?
(A vector $v \in T_p\mathbb{R}^2$ lies in $T_pS^1 \subset T_p\mathbb{R}^2$ iff it is tangent to S^1 at p i.e. perpendicular to the position vector p.)

(c) Let $f(x, y) = y$. Compute $v_i(f) = \frac{d}{dt}\Big|_{t=0} f(\gamma_i(t))$ for each i.

(d) Which curves define the same tangent vector in T_pS^1? Justify.

(e) In the angle chart θ with $\phi(\cos\theta, \sin\theta) = \theta$, write each tangent vector as a multiple of $\frac{\partial}{\partial\theta}\Big|_p$.

C2. (Tangent vectors on S^2 two charts.)

Let $p = (1, 0, 0) \in S^2$ and consider the two curves:

$$\gamma_1(t) = (\cos t, \sin t, 0), \qquad \gamma_2(t) = (\cos t, 0, \sin t).$$

(a) Compute the tangent vectors $v_1 = \gamma_1'(0)$ and $v_2 = \gamma_2'(0)$ as elements of $T_p\mathbb{R}^3$. Verify they are tangent to S^2 (perpendicular to p).

(b) In the spherical chart (θ, φ) with $(\theta, \varphi)(p) = (\pi/2, 0)$, express v_1 and v_2 as linear combinations of $\partial_\theta|_p$ and $\partial_\varphi|_p$.

(c) Now use the stereographic chart ψ_N from Exercise Set 6. Compute $\psi_N(p)$ and express v_1, v_2 as linear combinations of $\partial_u|_p$ and $\partial_v|_p$ in the stereographic chart.

(d) The two coordinate expressions for v_1 (in spherical vs stereographic) look different. Verify they are related by the Jacobian of the transition map $\psi_N \circ (\theta, \varphi)^{-1}$.

(e) **Key point.** The tangent vector v_1 is the same geometric object in both charts. Only its numerical components change. This is what it means for T_pM to be intrinsic to M.

C3. (The pushforward in coordinates.)

Let $f : S^1 \to S^2$ be the inclusion of the equator: $f(\cos\theta, \sin\theta) = (\cos\theta, \sin\theta, 0)$.

(a) Compute $df_p : T_pS^1 \to T_{f(p)}S^2$ at $p = (1, 0)$. What is the image of $\frac{\partial}{\partial\theta}\Big|_p$?

(b) Is df_p injective? Is f an immersion?

(c) Now consider $g : S^2 \to \mathbb{R}$ defined by $g(x, y, z) = z$. Compute $dg_p : T_pS^2 \to T_{g(p)}\mathbb{R} \cong \mathbb{R}$ at $p = (0, 0, 1)$ (north pole).

(d) At $p = (0, 0, 1)$, is $dg_p = 0$? What does this say about p as a critical point of g?

(e) Compute dg_p at $p = (1, 0, 0)$ (equator). Is it zero?

Vector fields in local coordinates

C4. (Vector fields on $\mathbb{R}^2$.)

For each of the following, write the vector field in the form $X = X^x\partial_x + X^y\partial_y$, compute its action on $f(x, y) = x^2 + y^2$, and find its zeros (points where $X = 0$).

(a) The rotation field: $X = -y\,\partial_x + x\,\partial_y$

(b) The dilation field: $X = x\,\partial_x + y\,\partial_y$

(c) The shear field: $X = y\,\partial_x$

(d) The constant field: $X = \partial_x + 2\partial_y$

C5. (Vector fields on S^1.)

In the angle chart θ on S^1, consider the following vector fields:

(a) $X = \frac{\partial}{\partial\theta}$: write its components in ambient (x, y) coordinates. Find its zeros. Is it nowhere-vanishing?

(b) $X = \sin\theta\,\frac{\partial}{\partial\theta}$: find its zeros. Is it smooth? Draw it.

(c) $X = |\sin\theta|\,\frac{\partial}{\partial\theta}$: is this smooth? Where does it fail to be smooth?

C6. (Vector fields on the cylinder $S^1 \times \mathbb{R}$.)

The cylinder $C = S^1 \times \mathbb{R}$ has coordinates (θ, z).

(a) Write three linearly independent vector fields on C: one purely angular, one purely vertical, one helical. Give each in the form $X = X^\theta\partial_\theta + X^z\partial_z$.

(b) For the helical field $X = \partial_\theta + \partial_z$, compute $X(f)$ for $f(\theta, z) = z\cos\theta$.

(c) Compute the Lie bracket $[\partial_\theta, \partial_z]$. What does this say about the flows of these two vector fields?

(d) Compute $[X, \partial_z]$ where $X = \sin\theta\,\partial_\theta + z\,\partial_z$.

C7. (Vector fields on S^2 the rotation fields.)

In ambient coordinates (x, y, z) on $S^2 \subset \mathbb{R}^3$, define:

$$X_1 = y\,\partial_z - z\,\partial_y, \qquad X_2 = z\,\partial_x - x\,\partial_z, \qquad X_3 = x\,\partial_y - y\,\partial_x.$$

(a) Verify that each X_i is tangent to S^2 at every point: show $X_i \cdot p = 0$ (the vector is perpendicular to the position vector $p = (x, y, z)$).

(b) Find the zeros of each X_i. How many zeros does each have?

(c) Compute the Lie brackets:

$$[X_1, X_2] = ?, \qquad [X_2, X_3] = ?, \qquad [X_3, X_1] = ?$$

(d) These are the generators of rotations about the x-, y-, z-axes. Compare with the commutation relations of angular momentum operators in quantum mechanics:

$$[L_x, L_y] = i\hbar L_z, \quad [L_y, L_z] = i\hbar L_x, \quad [L_z, L_x] = i\hbar L_y.$$

What is the relationship? What does the factor $i\hbar$ reflect?

(e) In the angle chart (θ, φ) on S^2, express X_3 in terms of ∂_θ and ∂_φ.

C8. (The tangent bundle of S^1.)

(a) Describe TS^1 as a set: what are its points?

(b) Show $TS^1 \cong S^1 \times \mathbb{R}$: define an explicit diffeomorphism using the angle chart.

(c) A nowhere-vanishing section $X : S^1 \to TS^1$ with $\pi \circ X = \mathrm{id}$ i.e. a nowhere-vanishing vector field exists on S^1. Give an explicit example.

(d) Explain why $TS^2 \ncong S^2 \times \mathbb{R}^2$ using the hairy ball theorem. Can you give a vector field on S^2 that vanishes at exactly two points?

Physics: Vector Fields as Equations of Motion

In Set 6 we learned that smooth structure gives us velocity. In this set we take the next step: the equations of motion *of a physical system are a vector field on the state space. Solving the equations of motion means finding the integral curves of this field. We also see the first instance of Noether's theorem that every symmetry vector field preserving the Lagrangian gives a conserved quantity. No Poisson brackets needed: the Lie derivative is enough.*

P1. (The pendulum as a vector field on TS^1.)

A simple pendulum of unit length and unit mass has configuration space S^1 (the angle θ). Its equation of motion is the second-order ODE:

$$\ddot{\theta} = -\sin\theta.$$

A second-order ODE on S^1 is equivalent to a first-order ODE on the **tangent bundle** $TS^1 \cong S^1 \times \mathbb{R}$, with coordinates (θ, ω) where $\omega = \dot{\theta}$ is the angular velocity.

(a) Rewrite the pendulum equation as a first-order system:

$$\dot{\theta} = ?, \qquad \dot{\omega} = ?$$

(b) This first-order system defines a vector field on TS^1:

$$X = \dot{\theta}\,\frac{\partial}{\partial\theta} + \dot{\omega}\,\frac{\partial}{\partial\omega}.$$

Write X explicitly in terms of θ and ω.

(c) Find all **fixed points** of X (points where $X = 0$). What do they correspond to physically?

(d) Describe the integral curves of X near each fixed point. Near $(\theta, \omega) = (0, 0)$: is the motion oscillatory or diverging? Near $(\theta, \omega) = (\pi, 0)$: is the fixed point stable?

(e) The **energy** of the pendulum is $E = \frac{1}{2}\omega^2 - \cos\theta$. Compute $X(E) = \mathcal{L}_X E$ (the Lie derivative of E along X, which equals $\frac{d}{dt}E(\gamma(t))$ along any integral curve γ). Show that $X(E) = 0$ the energy is conserved along every orbit.

(f) Draw the phase portrait of X on the (θ, ω) plane (i.e. $S^1 \times \mathbb{R}$). Mark:

- The two fixed points
- The level sets of E (these are the integral curves)
- The separatrix the curve $E = 1$ separating oscillating orbits from rotating orbits

(g) **Connection to the running problem.** In Sets 2–5, the dynamical system $f : X \to X$ was a discrete map. Now the dynamics is a continuous flow generated by a vector field. Both describe orbits, but the smooth structure lets us:

- Define velocity and acceleration at each instant
- Write the equations of motion as a differential equation
- Use conservation laws to understand the global orbit structure

The fixed points you found in (c) are the continuous-time version of the fixed points from Sets 2–3. The stability analysis from Set 6 (eigenvalues of df_{x^*}) now becomes the linearisation of X at the fixed point.

P2. (Noether's theorem via Lie derivatives.)

Setup. Let M be a smooth manifold (the configuration space) and $L : TM \to \mathbb{R}$ a smooth function (the Lagrangian). A vector field $X \in \mathfrak{X}(M)$ is a **symmetry** of L if

$$\mathcal{L}_X L = 0,$$

i.e. L does not change along the flow of X.

Noether's theorem (informal statement). Every symmetry vector field X of L gives rise to a quantity $Q_X : TM \to \mathbb{R}$ that is conserved along every solution of the Euler-Lagrange equations.

The conserved quantity is:

$$Q_X = \sum_{i=1}^{d} X^i(q) \frac{\partial L}{\partial \dot{q}^i},$$

where $X = \sum_i X^i(q)\, \partial_{q^i}$ in local coordinates (q^i) on M.

(a) **(Translation symmetry $\Rightarrow$ momentum conservation.)**
Let $M = \mathbb{R}$ and $L = \frac{1}{2}\dot{q}^2 - V(q)$. The vector field $X = \partial_q$ (constant translation) is a symmetry iff $\mathcal{L}_X L = 0$.
Compute $\mathcal{L}_X L = X(L)$ does it vanish for all V? For which V is X a symmetry?
Write the conserved quantity Q_X. What is it physically?

(b) **(Rotation symmetry $\Rightarrow$ angular momentum conservation.)**
Let $M = \mathbb{R}^2$ with coordinates (x, y) and $L = \frac{1}{2}(\dot{x}^2 + \dot{y}^2) - V(\sqrt{x^2 + y^2})$ (a particle in a central potential).
The rotation field $X_3 = x\partial_y - y\partial_x$ from C7 is a symmetry.

- Compute $\mathcal{L}_{X_3} L = X_3(L)$. Show it vanishes because V depends only on $r = \sqrt{x^2 + y^2}$.
- Compute the conserved quantity:

$$Q_{X_3} = X_3^x \frac{\partial L}{\partial \dot{x}} + X_3^y \frac{\partial L}{\partial \dot{y}}.$$

  What is it? (Write it in terms of $x, y, \dot{x}, \dot{y}$.)

(c) **(Three rotation symmetries on S^2.)**
Let $M = S^2$ with the Lagrangian of a free particle: $L = \frac{1}{2}(\dot{\theta}^2 + \sin^2\theta\, \dot{\varphi}^2)$ in spherical coordinates.
The three rotation fields X_1, X_2, X_3 from C7 are all symmetries of L.

- Verify $\mathcal{L}_{X_3} L = 0$ using $X_3 = \partial_\varphi$ in spherical coordinates.
  (In spherical coordinates, $X_3 = \partial_\varphi$ since it generates rotation about the z-axis.)
- Compute the conserved quantity Q_{X_3}. What is it physically?
- The three conserved quantities $Q_{X_1}, Q_{X_2}, Q_{X_3}$ are the three components of angular momentum. Their Lie brackets satisfy $[X_1, X_2] = X_3$ etc. (from

C7). Write one sentence connecting the algebra of symmetry vector fields to the algebra of conserved quantities.

(d) **(Time translation ⇒ energy conservation.)**

Time translation is a symmetry of any autonomous Lagrangian (one with no explicit time dependence). The corresponding conserved quantity is the energy:

$$E = \sum_i \dot{q}^i \frac{\partial L}{\partial \dot{q}^i} - L.$$

For the pendulum $L = \frac{1}{2}\dot{\theta}^2 - (1 - \cos\theta)$: compute E explicitly. Verify it matches the energy from P1(e).

(e) **(The structure of Noether's theorem.)** Complete the following table:

Symmetry vector field X	$\mathcal{L}_X L = 0$ means	Conserved quantity Q_X
∂_x (translation in x)	L independent of x	$p_x = \partial L/\partial \dot{x}$
$x\partial_y - y\partial_x$ (rotation)	L independent of φ	$L_z = x\dot{y} - y\dot{x}$
∂_t (time translation)	L independent of t	$E = \sum \dot{q}^i p_i - L$
X (general)	_____	_____

Write Noether's theorem in one sentence using only the language of vector fields and Lie derivatives no Lagrangian mechanics jargon.

The Lie bracket $[X_i, X_j]$ of symmetry fields is again a symmetry field the symmetries form a Lie algebra. This is the beginning of the story of Lie groups in physics, where continuous symmetry groups and their infinitesimal generators (Lie algebras) organise the conservation laws of every physical theory.

What structure is still missing? We can describe motion and symmetry, but we have no way to measure lengths or angles on M no notion of distance between configurations, no way to say which path is shortest. For that we need a **Riemannian metric** on the configuration manifold.***Reflection.*** *The equations of motion are a vector field on the tangent bundle TM. Solving them means finding integral curves. Symmetries are vector fields X satisfying $\mathcal{L}_X L = 0$, and each gives a conserved quantity via Noether's theorem.*

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